PER SCHOLAS

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Skill-Based Assignment:
Data Analysis and Visualization with Python

Objective:

In this skill-based assignment (SBA), you are expected to analyze and visualize the given dataset using Python, Pandas, Matplotlib, and other supporting libraries.

Time/Duration:

- You have **3 hours** to complete this SBA.
- Please ensure that you are not interrupted during the test.

Passing Criteria

- **Total Points: 100.**
- **Passing Score: 70% (70 points).**

Instructions

Section One

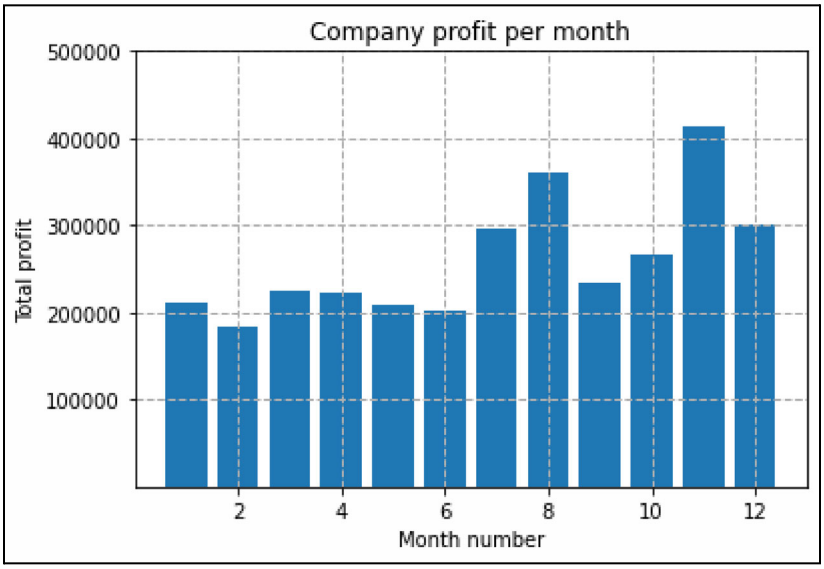
Use the following CSV file for this SBA:

➔ Click to download Sample Data. "woocommerce-product-export.csv"

Below are the Tasks:

1. Read Data from the woocommerce-product-export.csv file using the python library.
2. Show a concise summary of the columns using the info() method.
3. Show a summary of statistics pertaining to the columns.
4. Print the first five rows by default.
5. Print the last five rows by default.
6. Print the ***"total_profit"*** and ***"month_number"*** columns only.
7. **Read the total profit of all months and show it using the Bar plot.**
   - ➢ The generated **Bar plot** must include the following properties:
     - ❖ X label name = Month number.
     - ❖ Y label name = Total profit.
     - ❖ Title = Company profit per month.

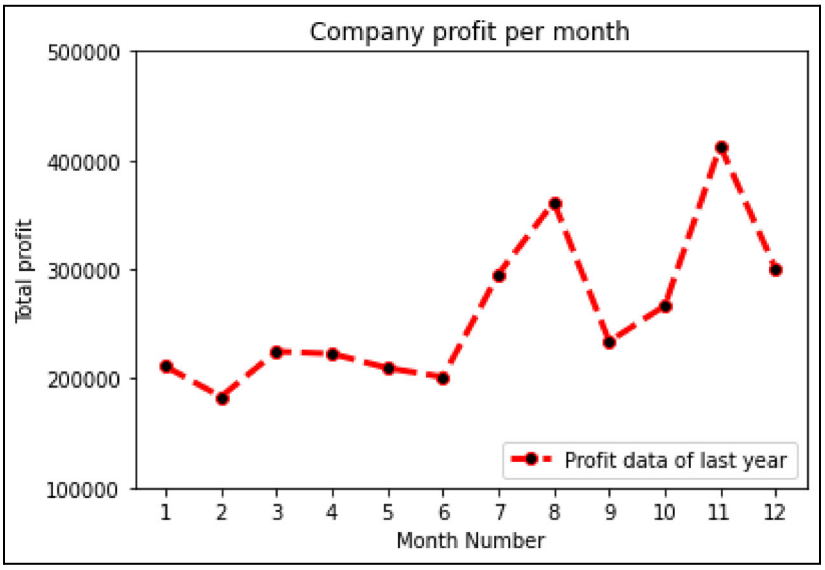
   The graph should look like the screenshot below:



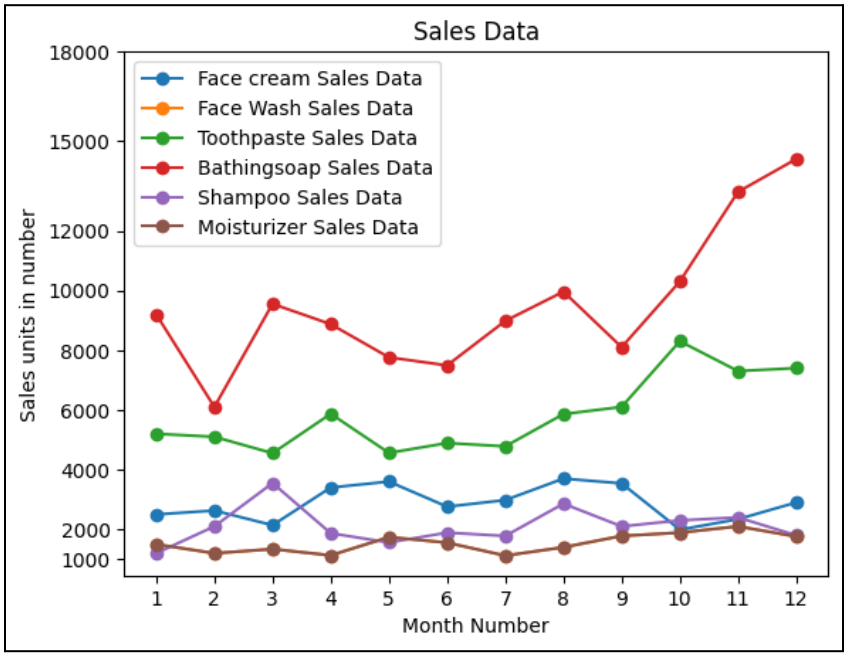
8. **Read the total profit of all of the months, and show the line plot with the following style properties:**
   - ➢ The generated **line plot** must include the following Style properties:

- ❖ The line style should be **dotted** and the **line color** should be **red**.
- ❖ Show the legend at the lower-right location.
- ❖ X label name = Month number
- ❖ Y label name = Total profit.
- ❖ Title = Company profit per month.
- ❖ Add a circle marker; the line marker color = black.
- ❖ The line width should be 3.

The graph should look like the screenshot below:

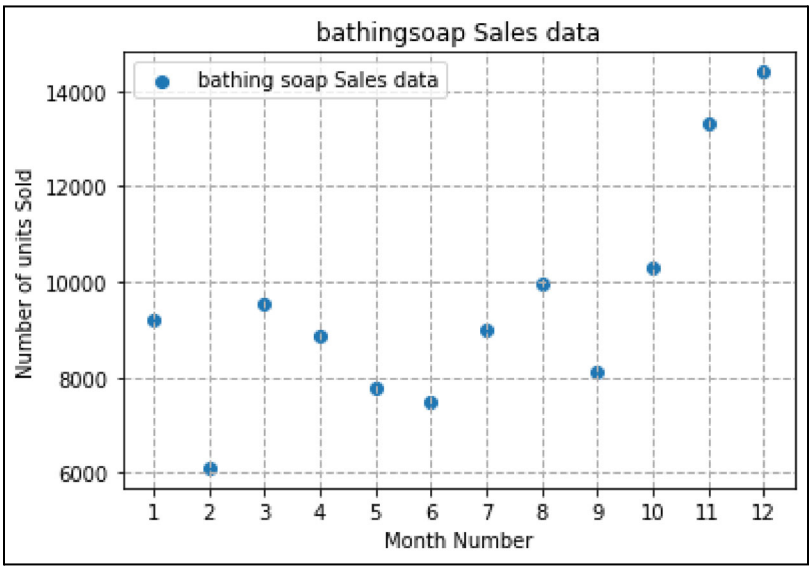


9. **Print all of the product sales data and show it using a multi-line plot.**
   - ➢ Display the number of units sold per month for each product using multi-line plots. (i.e., show a separate plotline for each product).
   - ➢ The graph should look like the screenshot below:



10. Read "*bathingsoap*" sales data for each month and show it using a scatter plot.

- ➢ Add a grid to the plot. The grid line style should be "–".
- ➢ The scatter plot should look like the screenshot below:



Section Two

Below are the tasks:

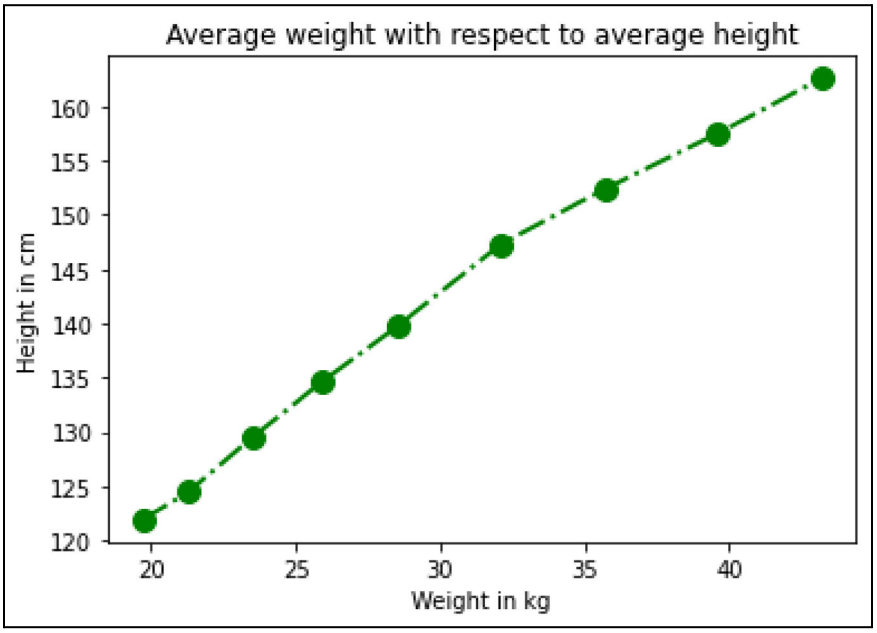
1. **Given Dataset:**
   date=["25/12","26/12","27/12"]
   temp=[8.5,10.5,6.8]

   A **line** chart of date versus temperature must include the following properties:
   - ➢ X label name = Date
   - ➢ Y label name = temperature
   - ➢ Title = Date-wise Temperature

2. **Given Dataset:**
   height=[121.9,124.5,129.5,134.6,139.7,147.3,152.4,157.5,162.6]
   weight=[19.7,21.3,23.5,25.9,28.5,32.1,35.7,39.6,43.2]

   Show the average weight against the average height (use a line chart or any other chart):
   - ➢ The generated **line plot** must include the following Style properties:
     - ❖ The line style should be dash-dot and the line color should be green.
     - ❖ Show the legend at the lower-right location.
     - ❖ X label name = Weight in kg.
     - ❖ Y label name = Height in cm.
     - ❖ Title = Average weight with respect to average height.
     - ❖ Add a circle marker: marker size = 10; circle marker color = green.

The graph should look like the screenshot below:



Grading Summary

Category	Points
Section One	65
Section Two	25
Project Submission	10
Total	**100 Points**

Scoring Summary

Level	Score Range	Descriptor
Excellent	90–100	All plots and code steps are accurate, well-documented, and fully meet formatting/style requirements.
Proficient	80–89	Most outputs are accurate with minor formatting or style errors. Documentation is clear.
Satisfactory	70–79	Meets minimum technical criteria; minor omissions or visual inaccuracies present.
Fail	Below 70	Missing required plots, incorrect outputs, poor documentation, or incomplete submission.

SBA submission guidelines

- Submit either an `.ipynb` **(Jupyter Notebook)** file or a `.py` **(Python script)** file.
    - The file must be named: `YOURNAME_386_SBA`.
    - Ensure that each step is **properly documented** with comments.

- Create a Word document.
    - Name the word document: `YOURNAME_386_DOC_SBA.doc.`
    - Take a screenshot of every result or chart.
    - Include all screenshots in a Word document.
    - Put any necessary information and details in the document, if needed.
    - Upload the Word document to Canvas.